GIVEN TABLE (UNNORMALIZED)

StdID	StdName	CourseID	CourseLength	UnitCode(s)	UnitName(s)	Lecturer(s)
001	Usman	A203	3	U45, U87	Database II, Programming	Ashraf Ghafoor
003	Nadeem	A104	4	U86, U45, U25	Algorithm, Database II, Business I	Naveed Ashraf
007	Abdullah	A203	3	U12, U46	Business II, Database I	Abid
010	Adnan	A323	2	U12, U86	Business II, Algorithm	Abid Naveed

STEP 1: UNF → 1NF

We break multi-valued fields (UnitCode, UnitName, Lecturer) into **atomic rows — 1 row per Unit per Student per Course**.

Table in 1NF:

StdID	StdName	CourseID	CourseLength	UnitCode	UnitName	Lecturer
001	Usman	A203	3	U45	Database II	Ashraf Ghafoor
001	Usman	A203	3	U87	Programming	Ashraf Ghafoor
003	Nadeem	A104	4	U86	Algorithm	Naveed Ashraf
003	Nadeem	A104	4	U45	Database II	Naveed Ashraf
003	Nadeem	A104	4	U25	Business I	Naveed Ashraf
007	Abdullah	A203	3	U12	Business II	Abid
007	Abdullah	A203	3	U46	Database I	Abid
010	Adnan	A323	2	U12	Business II	Abid Naveed
010	Adnan	A323	2	U86	Algorithm	Abid Naveed

Composite Primary Key:

StdID + UnitCode

Each row is one student's enrollment in one unit — that's what uniquely identifies a row.

STEP 2: FUNCTIONAL DEPENDENCIES (FDS)

From the 1NF data, we can derive:

1. StdID → StdName
2. CourseID → CourseLength
3. StdID → CourseID (Each student enrolled in one course)
4. UnitCode → UnitName
5. UnitCode → Lecturer *(seems true from table — same lecturer per Unit)*

🚨 **Partial Dependencies (Violating 2NF)**

Our composite PK is StdID + UnitCode, but:

- StdID → StdName, CourseID
- CourseID → CourseLength
- UnitCode → UnitName, Lecturer

These are all **partial dependencies**, so we must decompose to 2NF.

STEP 3: 2NF DECOMPOSITION

Table 1: Student(StdID, StdName, CourseID)

From FD: StdID → StdName, CourseID

Table 2: Course(CourseID, CourseLength)
From FD: CourseID → CourseLength

Table 3: Unit(UnitCode, UnitName, Lecturer)
From FD: UnitCode → UnitName, Lecturer

Table 4: Enrollment(StdID, UnitCode)
This is the **fact table** connecting Student and Unit — composite PK: StdID + UnitCode

All partial dependencies removed ✓
Now check for **transitive dependencies**

STEP 4: 3NF CHECK

We already have:

- All attributes in each table depend on the **full key**
- No attribute depends **on another non-key attribute**

So — there are **no transitive dependencies**!
🎉 Already in 3NF!

FINAL 3NF TABLES:

Student

StdID	StdName	CourseID
001	Usman	A203
003	Nadeem	A104
007	Abdullah	A203
010	Adnan	A323

Course

CourseID	CourseLength
A203	3
A104	4
A323	2

Unit

UnitCode	UnitName	Lecturer
U45	Database II	Ashraf Ghafoor
U87	Programming	Ashraf Ghafoor
U86	Algorithm	Naveed Ashraf
U25	Business I	Naveed Ashraf
U12	Business II	Abid / Abid Naveed*

U46	Database I	Abid

Note: U12 has two lecturers → if not a typo, this may violate FD. Confirm with instructor.

Enrollment

StdID	UnitCode
001	U45
001	U87
003	U86
003	U45
003	U25
007	U12
007	U46
010	U12
010	U86